STELLAE

GOVERNANCE INTELLIGENCE PLATFORM

INTERNAL DOCUMENT
ACADEMIC FOUNDATION
VERSION 2.0 — MAY 2026
CONFIDENTIAL

WHY STELLAE EXISTS

The Problem Is Documented. *The Solution Is Built.*

Stellae is not built on intuition. It is built on a body of peer-reviewed research spanning 70 years of megaproject data, 258 infrastructure projects across 20 nations, 85 international megaprojects studied with structural equation modeling, and an industry-wide productivity analysis of over 2,500 construction projects across 5 continents — from ADNOC, Aramco, Tsinghua University, the McKinsey Global Institute, and the University of Oxford. The academic foundation below explains precisely why capital project governance fails — and how Stellae addresses each failure mode.

9/10	44.7%	70yr	89%
MEGAPROJECTS WITH COST OVERRUN	AVERAGE RAIL COST OVERRUN (CONSTANT PRICES)	PERIOD STUDIED — ACCURACY HAS NOT IMPROVED	OF IM PERFORMANCE VARIANCE EXPLAINED BY FORMAL GOVERNANCE

57%	77%	2nd
OF CONSTRUCTION PROJECTS OVER BUDGET — MCKINSEY 2016	OF CONSTRUCTION PROJECTS DELAYED — MCKINSEY 2016	LOWEST LABOR PRODUCTIVITY OF ALL GLOBAL ECONOMIC SECTORS — CONSTRUCTION, AHEAD OF ONLY AGRICULTURE

THE ARGUMENT IN FOUR STEPS

PROBLEM — FLYVBJERG 2007

9 of 10 megaprojects fail on cost and schedule. The cause is not technical error — it is deliberate misrepresentation and lack of accountability structures.

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SCALE — MCKINSEY 2016

Construction has the 2nd lowest productivity of all global sectors. Fragmented document management is the primary root cause — not workforce quality.

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MECHANISM — SHEN ET AL. 2021

Formal governance is the dominant predictor of interface management performance, which directly drives project outcomes in cost, time, and quality.

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SOLUTION — STELLAE 2026

Real-time governance enforcement against Authority Matrix and Escalation Rules. Every decision audited. Every violation flagged. Accountability made automatic.

RESEARCH PILLARS

2020

SPE-203431-MS
ABU DHABI
ADNOC

✓ IMPLEMENTED

Al-Marzouqi et al. — Society of Petroleum Engineers

Micro-triggers for Early Risk Detection in EPC Projects

Identifies 25 specific micro-triggers across engineering, procurement, construction, controls, and closeout phases that signal governance failures before they escalate into cost overruns or schedule delays. Validated in ADNOC megaprojects in Abu Dhabi. Each trigger has documented causal links to specific project failure modes.

In Stellae today: All 25 micro-triggers are embedded in the system prompt. Every document analyzed is scanned for TBE pendiente, RFIs >72h sin respuesta, LLIs en ruta crítica, PTW recurrentes, FEED deficiencies, and 20 additional EPC-specific signals. Five advanced logic rules apply cross-document reasoning: LLI on critical path escalation, SLA contractual silence detection, PTW productivity loss estimation, FEED deficiency scope risk flagging, and 90-day frontend plan audit. Industry dictionaries (O&G, Mining, Infrastructure, Energy) ensure vocabulary precision per client sector.

2007

AALBORG UNIVERSITY OXFORD

✓ FOUNDATION

Bent Flyvbjerg — Aalborg University / University of Oxford

Megaproject Policy and Planning: Problems, Causes, Cures

Analysis of 258 transportation infrastructure projects across 20 nations over 70 years. Establishes with statistical significance that cost overruns are not caused by technical error or optimism bias, but by deliberate strategic misrepresentation. Average rail overrun: 44.7% (constant prices). Accuracy has not improved over 70 years without structural intervention. The solution requires accountability structures and independent governance oversight — not better forecasting methods.

In Stellae today: The Audit Trail is the accountability structure Flyvbjerg calls for. Every decision, every violation, every finding is timestamped and immutable. Project owners cannot retroactively claim "the team made errors" — the record exists. The Authority Matrix enforces who can approve what, eliminating the incentive to misrepresent. Stellae operationalizes Flyvbjerg's structural solution in real time, not in post-mortem reports.

2016

MCKINSEY GLOBAL INSTITUTE

✓ FOUNDATION

McKinsey Global Institute — Analysis of 2,500+ construction projects across 5 continents

Imagining Construction's Digital Future

Analysis of over 2,500 construction projects globally establishes that construction has the second lowest labor productivity of all economic sectors — ahead of only agriculture. Productivity growth over the past 20 years: 1% annually, vs. 2.8% for the global economy. 57% of projects are over budget; 77% are delayed. The root cause is not workforce quality — it is fragmented document management, lack of standardization, and absence of real-time information integration across project stakeholders. Digital tools that integrate information flows in real time are identified as the primary lever to close the productivity gap.

In Stellae today: Stellae directly addresses McKinsey's root cause: it integrates information flows across project documents in real time, replacing the fragmented spreadsheet and email chains McKinsey identifies as the primary source of the productivity gap. The hybrid document ingestion (PDF digital, scanned PDF with automatic OCR, DOCX, field reports) ensures no document type is excluded from the governance layer — exactly the standardization lever McKinsey prescribes. Every week a project runs without Stellae is a week operating at the second-lowest productivity level in the global economy.

2021

TSINGHUA UNIVERSITY MELBOURNE

✓ FOUNDATION

Shen, Tang, Wang, Duffield et al. — J. Construction Engineering Management

Managing Interfaces in Large-Scale Projects: The Roles of Formal Governance and Partnering

Empirical study of 85 international megaprojects across Africa, Asia, Europe, and South America. Structural equation modeling demonstrates that formal governance explains 89% of variance in interface management performance, which in turn has the strongest direct effect on project outcomes ($\beta=0.88$, $p<0.01$). Formal governance significantly outperforms informal partnering and boundary spanning activities as a predictor. The more complex the project, the stronger the effect.

In Stellae today: The AI Governance Generator, Authority Matrix, and Escalation Rules implement exactly the formal governance structure this study identifies as the dominant predictor of project success. Stellae does not rely on trust or informal coordination — it codifies the governance framework and enforces it automatically on every document analyzed. The 89% variance figure is not marketing language: it is the empirical justification for why Stellae's governance layer is the highest-leverage intervention available to a Project Director.

2022

SPE-222388-MS
SAUDI ARAMCO

STAGE 3 ROADMAP

Saudi Aramco — Society of Petroleum Engineers

Interface Agreements in Capital Projects: Preventing Scope Gaps

Documents how interface agreements between disciplines, contractors, and vendors prevent scope gaps and unplanned integration failures. Validated in Aramco capital projects. Interfaces without formal agreements are the primary source of late-stage design changes, rework, and cost escalation in large EPC environments. A single diameter change can affect 7 interface agreements simultaneously.

In Stellae Stage 3: Interface Impact Analysis module — Stellae will automatically identify when a document references a scope boundary between disciplines or vendors, flag missing interface agreements, and map cascade impacts. A 24" to 30" pipe diameter change detected in a minute will automatically trigger: "5 Interface Agreements affected — Structural, Instrumentation, Electrical, Civil, Regulatory."

2020

SPE-203215-MS
FEED PHASE

STAGE 2 ROADMAP

Society of Petroleum Engineers — Capital Projects FEED

Big-Ticket Decisions in FEED: How Early Choices Drive Final Cost

Establishes that 70–80% of total project cost is committed during the FEED phase through a small number of high-impact decisions. Late identification of these decisions is the primary cause of change orders and scope creep. A decision made in Construction to reverse a FEED choice costs 25x more than making the right choice during FEED. Identifies decision weight by project phase as a measurable governance metric.

In Stellae Stage 2: Decision Tracker module — each finding will be assigned decision weight by phase (FEED, detailed engineering, procurement, construction). An orphan decision open for 23 days in Construction will display: "Estimated cost of inaction: $287,500 (based on $50,000/day burn rate × 25x Construction phase multiplier)." Converts qualitative governance into quantitative risk scoring.

"The question is not whether formal governance works. The empirical evidence is unambiguous across 70 years, 258 projects, and 5 continents. The question is whether it exists in real time — or only in the post-mortem."

The research compiled here spans peer-reviewed journals (Journal of Construction Engineering and Management, Journal of the American Planning Association), Society of Petroleum Engineers conference papers validated at ADNOC and Aramco, McKinsey Global Institute industry analysis covering 2,500+ projects, and doctoral research validated by the UK Treasury and the American Planning Association.

Stellae is the first product to operationalize this body of knowledge in a real-time AI platform for capital project governance. The academic foundation is not marketing language — it is the specification from which the system was built.

Dennis Malave — Founder
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